Fiscal Year:	FY 2020	**Task Last Updated:**	FY 05/31/2020
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Project Title:	Microgravity Effects on Co-cultured Vascular Cells Types		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Cell Culture		
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Comments:			
Project Type:	Ground,NASA GeneLab	**Solicitation / Funding Source:**	2016 Space Biology (ROSBio) NNH16ZTT001N-GeneLab. Appendix A: Translational Systems Biology and Informatics Research Using the GeneLab Data System
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Task Description:	Upon returning home from space flight, the cardiovascular system of astronauts is altered, with changes manifesting in what is termed cardiovascular deconditioning. Essentially, this means that their cardiovascular system is not functioning properly. It is well known that proper function of the cardiovascular system is largely dependent on the maintenance of healthy functioning vascular cells, namely endothelial cells that line blood vessels, smooth muscle cells which make up the underlying vessel wall, and a population of “repair” cells termed endothelial stem/progenitor cells found in circulating blood. Work has shown that endothelial cells are affected by microgravity conditions. In fact, in a groundbreaking study provided through GeneLab, endothelial cells were studied under conditions of space flight. The study termed Spaceflight of Human Umbilical Vein Endothelial Cells (SPHINX), was part of the Progress 40P space mission which lasted for 10 days. The results obtained from GeneLab on the space study as well as published reports show that in response to real microgravity as seen in space, endothelial cells become altered and dysfunctional. This study evaluated, via microarray technology, gene expression changes that govern a myriad of functional processes. It is our hypothesis that these changes in endothelial cells will significantly alter important cell-cell communications between endothelial cells and smooth muscle cells as well as endothelial cells and circulating progenitor cells. Given the critical nature of these cell-cell interactions in maintaining vessel homeostasis, it is important that we understand the effect of microgravity on this cross talk, and correlate our findings with the gene and protein expression changes in endothelial cells as shown in the SPHINX study. This proposal seeks to study co-cultured endothelial and smooth muscle cells under modelled microgravity to assess changes in smooth muscle cell processes, such as proliferation and contraction. Similarly, we seek to study co-cultured endothelial cells and circulating stem/progenitor cells under conditions of modelled microgravity to assess changes in the stem cells differentiation and anti-thrombogenic function. We also will assess functional changes in the endothelial cells themselves to correlate the genetic changes with anti-thrombogenic responses. The work proposed is inspired by the GeneLab data from the SPHINX study that revealed important genetic changes in endothelial cells. We hope to expand on this data by revealing important functional behaviors of endothelial cells towards the goal of the development of cell based countermeasures to address the challenges of cardiovascular deconditioning.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In a groundbreaking study provided through GeneLab, endothelial cells were studied under conditions of space flight. The study termed Spaceflight of Human Umbilical Vein Endothelial Cells (SPHINX) was part of the Progress 40P space mission. Using this work and our own previous findings as a foundation, we seek to understand the impact of microgravity on cell-cell communication between the cells of the vasculature, endothelial, and smooth muscle cells. Given the critical nature of these cell-cell interactions in maintaining vessel homeostasis, it is important that we understand the effect of microgravity on this cross talk, and correlate our findings with the gene and protein expression changes in endothelial cells as shown in the SPHINX study. Our research is motivated by the fact that dysfunction of these cells has been implicated in the onset and progression of cardiovascular disease. Once an astronaut who completed time in space returns to the normal gravity of Earth, it is our goal that the work we conducted will reveal the effect of the lack of gravity on the astronaut's overall vascular health. Our work has impacts for life on Earth and in space. We are modelling our experiments to mimic the conditions in space where there is a lack of gravity; however, we are also including control samples that are treated under normal gravity. Our work will reveal important changes in cell-cell communication as a result of altered environmental conditions, which could lead to new therapeutic targets in the treatment of cardiovascular disease.
Task Progress:	Based upon a rich body of work describing vascular cell dysfunction in microgravity, we sought to investigate the important cell-cell communications between cells that make up the vascular wall, endothelial and smooth muscle cells. Our central hypothesis is that microgravity alters cell-cell communication between vascular endothelial cells and other cells of the vasculature, specifically smooth muscle cells. One of our main goals was to evaluate how the function of endothelial cells (EC) changes in conditions of microgravity. For this work we are interested in proteins secreted by endothelial cells that are responsible for the health and function of the underlying smooth muscle cells. In this work, our samples are cultured both in microgravity and in normal gravity and changes in the cells' gene expression was assessed as a result of microgravity. This work represents the secretory function of the endothelium, a critical function that gives ECs the ability to regulate the health of the vasculature. We analyzed 4 genes of interest and we show that relative to day 0, the four genes of interest were increased. Overall, these results are promising and demonstrate that microgravity results in upregulation of gene responsible for the secretion of important molecules that regulate smooth muscle cell (SMC) function. In addition to gene expression, we also characterized protein expression. We assessed the levels of proteins of interest from endothelial cells cultured in both normal gravity as well as microgravity over 21 days. Our data show that under conditions of microgravity, similar to gene expression, we see protein expression and secretion levels are also increased in microgravity. Collectively, this data provides a picture for the dysregulation of endothelial cells in microgravity at the gene and protein levels. Another goal was to evaluate the changes in endothelial cells and smooth muscle cells in co-culture with each other under conditions of simulated microgravity. To achieve this, we were challenged to develop a co-culture system that would allow for both populations to be grown in contact with each other but still allow them to be separated and analyzed independently. Using magnetic cell separation technology, we were able to optimize a bead-based method and isolate each population of cells intact. Our optimizations were done to ensure best possible cell survival, proliferation, and downstream biological response following long term microgravity culture. Once optimized we characterized the changes that endothelial and smooth muscle cells undergo when cultured in contact with each other in co-culture and in microgravity. Our interests were in functional deviations that are manifest in secretory, gene, and protein expression. We co-cultured ECs and SMCs in microgravity and showed microgravity induced changes in gene expression of both cell types. The gene expression analysis can be grouped into structural genes and those related to cell-cell adhesion. Structurally we show that when endothelial and smooth muscle cells are co-cultured structural genes are increased in microgravity. This trend is consistent for both normal vs microgravity and is greater in microgravity vs. normal gravity. The shift in structural gene expression is indicative of a shift from the proliferative to the contractile phenotype of SMCs. As it relates to genes involved in cell-cell communication, we show a reduction in some genes while others remained unchanged in co-culture and in microgravity, which is somewhat in agreement with our previous studies. This trend provides insight into the integrity of the endothelium, an important function. This is in contrast to what we saw with smooth muscle cells, which showed significant decreases in cell adhesion related genes, such as focal adhesion kinase, when in microgravity. Collectively these data provide some insight into the phenotypic modulation of SMC and the membrane integrity of ECs and perhaps the EC function of leukocyte recruitment. Indeed, our study shows

	dysfunction of endothelial cells and smooth muscle cells when cultured in microgravity together. Some of the changes we saw were consistent with endothelial cell-smooth muscle cell co-culture, but some were an alterations due to microgravity. Overall, our accomplishments included the understanding of the role of simulated microgravity in the dysfunction of vascular endothelial and smooth muscle cell when in co-culture. Specifically, we report on the changes in gene and protein expression, and correlate those changes to unhealthy vascular function. This work provides some insight into the observed cardiovascular deconditioning seen during and post space flight.
Bibliography Type:	Description: (Last Updated: 01/31/2022)
Abstracts for Journals and Proceedings	Scotti M, Allen JB. "Changes in vascular cell function under conditions of microgravity." Presented at the American Chemical Society National Meeting, New Orlando, FL March 31-April 4, 2019. Abstracts. American Chemical Society National Meeting, New Orlando, FL March 31-April 4, 2019. , Mar-2019